



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

A.C.E.S – Autonomous Cyber-defence and Enforcement System

**The domain of the Project:
Cybersecurity**

Team Mentors (and their designation):
Mr. Derick Johnson (Cybersecurity Trainer, Sure ProEd)

Team Members:
Mr. Rakesh S

Period of the project
February 2026 to August 2026



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Declaration

The project titled “Autonomous Cyber-defence & Enforcement System” has been mentored by Mr. Derick Johnson, organised by SURE Trust, from February 2026 to August 2026, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Team Members:

Mr. Rakesh S – 

Mr. Derick Johnson
Cybersecurity Trainer—Sure ProEd

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Executive Director & Founder
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Executive Summary

A.C.E.S. (Autonomous Cyber-defence & Enforcement System) is a production-grade Zero Trust security platform built entirely in Python. It simulates a corporate financial portal — *NorthStar Capital* — defended by three simultaneous and independent detection layers. The first, **PortTrap**, is a multi-port TCP honeypot that catches port scans and reconnaissance probes, responding with an instant IP ban in under 5ms and immediate session revocation. The second, **HoneytokenWatcher**, is a filesystem monitor that detects credential file access and insider threats, triggering the full SOAR pipeline along with a SOC alert and ban. The third, **BehaviorScorer**, is an IsolationForest-based machine learning layer that flags anomalous click speed and request rates, raising a SOC alert once the anomaly score crosses 0.50 and auto-containing the session once it reaches 0.85.

All three layers feed into a unified **SOC analyst dashboard** that streams live threat events over WebSocket with sub-50ms latency. The system autonomously bans attacker IPs, revokes active sessions, and logs structured forensic records — while keeping the analyst in full control through a triage queue and manual override controls.

The anomaly detection model was trained on **real benign network flows** from the CIC-IoT-2023 dataset. Validated against live Kali Linux attack scenarios, the system correctly detected and contained every simulated threat: port scans, directory brute-force, honeytoken access, and anomalous session behaviour.



Introduction

Background and Context

Modern enterprise security has moved beyond perimeter defence. The Zero Trust model — *"never trust, always verify"* — treats every request as potentially hostile regardless of network location. Implementing this at the application layer requires:

- Continuous behavioural monitoring of every session
- Automated threat response that does not wait for human intervention
- Real-time analyst visibility into the full attack surface

Most small and medium organisations cannot afford commercial SIEM/SOAR platforms (Splunk, Palo Alto Cortex XSOAR) which cost upward of \$100,000/year. A.C.E.S. demonstrates that the same core capabilities can be built with open-source components at zero licensing cost.

Problem Statement

How can a system simultaneously detect external network attackers, filesystem-level intruders, and insider threats — and respond to all three automatically, in real time, without requiring a security engineer to watch a terminal?

Scope

The project builds the complete detection and response stack:

- ML model training on real network flow data
- A scored HTTP financial portal (the target surface)
- Three-layer deception and detection infrastructure
- An event-driven SOAR pipeline
- A five-page live SOC analyst dashboard

Validated against real Kali Linux attack scenarios on a local network.



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Limitations

- The portal simulates financial transactions rather than implementing real banking logic
- AbuseIPDB reporting runs in offline-mock mode unless a live API key is configured
- Honeypot ports below 1024 require Administrator privileges on Windows

Innovation Component

Most academic security projects implement one detection mechanism in isolation — either a honeypot *or* ML anomaly detection *or* a dashboard. A.C.E.S. integrates three orthogonal detection layers into a single real-time pipeline with a unified analyst interface — the same architecture used in commercial SIEM/SOAR products. The system is also unique in using a model trained on real network traffic (not synthetic data) and validating it against live attack tools.



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Project Objectives

The project set out to achieve seven concrete objectives, each tied to a measurable success criterion. The first was to train an IsolationForest model on real benign network flows, with success defined as a clear score separation between benign traffic (0.05–0.25) and attack traffic (0.70–1.0). The second was to score every portal session in real time, such that the score updates on every click within a 100ms round-trip. The third was to auto-escalate any session whose score reached 0.85 or higher, with the SOAR pipeline banning the IP and revoking the session within 5ms. The fourth was to deploy multi-port TCP honeypots on ports 22, 445, and 3389, so that any TCP connect attempt would trigger an instant ban and SOC alert. The fifth was to deploy a filesystem honeypot watched by a background daemon, such that any access to it would trigger the full SOAR pipeline. The sixth was to stream all threat events to the SOC dashboard over WebSocket, with alerts appearing on the dashboard within 50ms of being triggered. The seventh and final objective was to provide five analytical views on the SOC dashboard, all functional and role-gated to the SOC engineer.

Expected Deliverables:

train_behavior_model.py ← offline model trainer
run.py ← production server bootstrap
data/behavior_model.pkl ← trained IsolationForest + RobustScaler
data/aces_production.db ← SQLite event log database
src/
zero_trust/scorer.py ← BehaviorScorer
zero_trust/models.py ← data models
soc_pipeline/soar_engine.py
soc_pipeline/ws_dashboard.py
soc_pipeline/event_hub.py
deception/port_trap.py
deception/token_watcher.py
portal/financial_shell.py
portal/static/soc.html ← SOC dashboard
portal/static/portal.html ← financial portal
portal/static/index.html ← landing / login



Methodology and Results

Methods and Technology

Each component of the stack was chosen deliberately. Anomaly detection uses scikit-learn's IsolationForest, selected because it is unsupervised — it requires no labelled attack data — and offers $O(n \log n)$ inference. Feature scaling uses RobustScaler, which handles extreme outliers in network flow distributions better than StandardScaler. The async HTTP server is built on Python's asyncio, so a single event loop can handle thousands of concurrent connections without threads. WebSocket support was implemented manually against RFC 6455 to avoid a heavy framework dependency and retain full control over framing and keepalive. The database is SQLite running in WAL mode, giving in-process, zero-latency storage with concurrent reads that survives restarts. Filesystem watching uses the watchdog library, a cross-platform inotify/FSEvents wrapper, and the frontend charts are built with Chart.js 4.4 for the KDE density plots and live time-series streaming.

Dataset

The model was trained on CIC-IoT-2023 — BenignTraffic.pcap.csv, sourced from the Canadian Institute for Cybersecurity. The dataset contains 362,361 rows, from which 29 features were used — covering rate, inter-arrival time (IAT), variance, TCP flags, protocol presence, and packet size statistics. Preprocessing replaced $\pm\text{Inf}/\text{NaN}$ values, dropped the affected rows, and applied RobustScaler. The full dataset was used for training, with 10,000 held-out rows reserved for validation.

ML Scoring Formula

```
raw_score = IsolationForest.decision_function(X) # negative = anomalous
clamped   = clip(raw_score, -0.5, +0.5)
score     = 0.5 - clamped                       # maps to [0, 1]
```

score = 0.0 → deep inside benign cluster (perfectly normal)

score = 0.50 → SOC alert threshold (analyst review)

score = 0.85 → SOAR auto-containment (ban + session revoke)

score = 1.0 → maximum anomaly



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Validation Results

The system was validated against seven test scenarios, and every one produced the expected result. A full TCP connect port scan (nmap -sT -p 3389) resulted in the IP being banned and a PORT_TRAP alert on the SOC dashboard. A fast connect scan (nc -z 10.x.x.x 3389) triggered a ban within 5ms. A directory brute-force attempt against a bait path (curl /wp-admin) produced an instant ban and a HONEYTOKEN_HIT alert. Clicking the db_config.json credential file in the portal triggered the SOAR pipeline and revoked the session. Normal portal browsing at a low click rate scored between 0.05 and 0.20 and correctly produced no alert, confirming no false positives. Anomalous rapid clicking pushed the score above 0.50 and raised a SOC alert. Finally, an HTTP request made after a ban (curl http://host:8000/) was correctly blocked with a 403 quarantine page.

Dashboard Pages

Page 1 — Active Logs & Details Live terminal-style event stream over WebSocket. Every PORT_TRAP, HONEYTOKEN_ACCESS, and AI_ANOMALY event appears within 50ms of firing. Bottom panel shows the Top 5 most volatile IPs by anomaly score.

Page 2 — Live Triage Queue Table of all security events with unread badge counter. Clicking a row reveals the full event detail panel (IP, username, role, trigger, score, timestamp) and provides one-click analyst actions: Investigate, Manual Ban, Remove Ban, Report to AbuseIPDB.

Page 3 — Deception & Honeytokens Shows live honeytoken file content, active honeypot port status, honeytoken trip history, SOAR mode toggle (Monitor-Only vs Enforce), and a live honeypot management control for changing trap ports without restarting the server.

Page 4 — Anomaly Score Distribution KDE density plot showing green benign traffic curve (peaks at 0.10–0.15) vs red attack traffic curve (peaks at 0.80–0.95), scored by the production IsolationForest against real CSV rows and synthetic scanner profiles. Two dashed threshold lines at 0.50 and 0.85.

Page 5 — Baseline Timeline Live time-series chart that polls the scorer every 5 seconds and plots anomaly score vs the 0.50 baseline threshold. Score line turns red when crossing into the anomaly zone. Slider controls adjust request rate and click interval to simulate different traffic profiles in real time.

GitHub Link: [Project repository](#)



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Social / Industry Relevance

Industry Relevance

Zero Trust architecture is the current global standard for enterprise security — mandated by the US Executive Order on Cybersecurity (May 2021) and adopted by Google (BeyondCorp), Microsoft (Azure AD Conditional Access), and Cloudflare (Zero Trust Network Access). The IsolationForest approach to anomaly detection is used in production by Netflix (infrastructure monitoring), Amazon (fraud detection), and multiple cybersecurity vendors.

A.C.E.S. implements the same core capabilities as commercial platforms: its EventHub and SOC Dashboard mirror the role of a Splunk SIEM, its SOAREngine mirrors Palo Alto Cortex XSOAR, its BehaviorScorer and IsolationForest layer mirrors CrowdStrike Falcon, and its PortTrap and HoneytokenWatcher components mirror the honeypot capabilities offered by vendors such as Attivo Networks.

Social Relevance

Cybercrime costs the global economy over \$8 trillion annually (Cybersecurity Ventures, 2023). Small and medium businesses — which disproportionately serve rural and underserved communities — are the most frequent targets because they cannot afford enterprise security tools.

A.C.E.S. demonstrates that production-quality threat detection can be built entirely with open-source components at zero licensing cost, making it accessible to hospitals, NGOs, local government bodies, and educational institutions that operate without security budgets.



Learning and Reflection

Technical Learnings

Machine Learning

- How IsolationForest works: isolation path length as anomaly signal; why shorter average path = more anomalous
- Why contamination=0.02 (2%) gives a tighter boundary than the default 0.10
- Why RobustScaler (median/IQR) outperforms StandardScaler (mean/std) for network flow data with long-tailed distributions
- The difference between unsupervised anomaly detection and supervised classification — and why unsupervised is the only option when you have no labelled attack data

Async Python

- How the asyncio event loop works: cooperative multitasking, why await does not block, how create_task enables concurrency
- The difference between asyncio.Queue fan-out and a single shared queue — and why fan-out is required for independent consumers at different speeds
- How to bridge OS-level threads (watchdog file monitor) into the async event loop using call_soon_threadsafe

Network Security

- Why Nmap's default -sS SYN scan does not trigger asyncio.start_server but -sT full connect does — and how this gap affects real-world honeypot effectiveness
- How TCP banner responses reveal service versions to attackers — and why sending zero bytes is the correct honeypot design
- What TTL values, TCP flag patterns, and IAT distributions reveal about traffic type

System Design

- Why the ban is written to SQLite *before* SOAR fires — to prevent a race window where the next HTTP request arrives before the async task executes
- Why WAL (Write-Ahead Logging) mode allows concurrent SQLite readers without blocking the writer
- The role of idempotency in security operations — why ban_ip() uses INSERT OR REPLACE so repeated calls are safe



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Overall Experience

Building A.C.E.S. from scratch gave me an end-to-end understanding of how production security systems are actually constructed — not just the theory but the exact code paths: from a TCP packet arriving on a honeypot port, to an alert appearing on a SOC analyst's screen 50ms later, to an IP being permanently blocked in the database.

The most valuable insight was that every architectural decision in a security system has a direct threat model consequence. Why does the EventHub use per-subscriber queues? Because if the analytics consumer blocks, it must not delay the WebSocket broadcast that the analyst is watching. Why is the honeypot file pre-populated with realistic-looking fake credentials? Because an attacker who opens an empty file knows immediately it is a trap.



Conclusion and Future Scope

Recap of Objectives and Achievements

Every objective set out at the start of the project was completed. The IsolationForest model was trained on 362,000 real flows and achieved clear score separation. Real-time session scoring was implemented with sub-100ms latency, scoring every click. Auto-containment at a score of 0.85 or higher was achieved, with bans and revocations completed in under 5ms. Multi-port TCP honeypots were deployed on ports 22, 445, and 3389. The filesystem honeypot watcher was completed, with any access triggering the SOAR pipeline. The live SOC dashboard was delivered with all five views streaming over WebSocket. Finally, the system was validated against Kali Linux attacks, with every scenario successfully detected.

Future Scope

Several enhancements are planned for future iterations of the system. Adding HTTPS/TLS via Let's Encrypt would prevent credential interception in a production deployment. Migrating to a PostgreSQL backend would enable multi-analyst concurrent access along with long-term log retention and querying. A supervised ML layer trained on CIC-IDS-2018 would add labelled attack classification alongside the current unsupervised baseline. Adding PCAP capture on the honeypot ports would provide full packet-level forensic evidence for incident response. Enabling live AbuseIPDB reporting would automate community threat intelligence sharing. Packaging the system with Docker / docker-compose would allow one-command reproducible deployment. Email/SMS alerting would notify the analyst of critical events even when they are not connected to the dashboard. GeoIP integration would map attacker IPs to geographic locations on the SOC dashboard. Finally, multi-tenant support would allow separate event streams per organisation on a shared platform.